Consistency Is Not Enough: From Routing Stability to Compositional Robustness in Mixture-of-Experts

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Abstract

Fine-tuning a mixture-of-experts (MoE) router to reuse a stable set of experts across a sequence is usually motivated as a serving-efficiency trick, but stable routing is only useful insofar as it reflects something deeper: that individual experts specialize into reusable, semantically coherent modules, and that this modularity in turn makes the model's behavior more robust when its input changes. We make this chain explicit and test each link. Using CAR-MoE (Cache-Aware Routing for MoE), a GRPO-trained router objective that rewards a simulated working-set hit ratio, we measure, on top of two backbones (Phi-tiny-MoE-instruct, OLMoE-1B-7B) and five objectives (base, supervised fine-tuning, CAR-MoE, an option-critic-style controller, and MELINOE): (i) routing *consistency* under meaning-preserving surface perturbations of the input; (ii) *modularity*, via each expert's specialization to one of two content domains; and (iii) downstream *robustness*, via the accuracy drop on GSM8K under the same perturbation. We find that consistency and modularity track robustness within the family of objectives that route non-degenerately, and that CAR-MoE achieves the smallest accuracy drop of any objective on *both* backbones – beating the untuned base model outright on OLMoE-1B-7B – without any accompanying spike in aggregate consistency or specialization statistics. The option-critic controller, in contrast, is a sharp counterexample: it reaches near-perfect routing consistency and the highest specialization score of any objective by collapsing its routing onto a handful of experts, and this degenerate "modularity" coincides with the *worst* robustness on Phi-tiny-MoE. Stability, we conclude, is necessary but not sufficient evidence of useful modularity; it must be checked against routing diversity and downstream robustness directly, not assumed.

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1 Introduction

A mixture-of-experts router fine-tuned to reuse a small, stable set of experts per sequence is typically justified purely as a memory-traffic optimization. But routing consistency, on its own, is not obviously a virtue – a router that always activates the same expert regardless of input is maximally consistent and maximally useless. What should actually matter is *modularity*: whether individual experts specialize into coherent, reusable functional units, so a consistent routing decision reflects a genuine correspondence between content and expert identity rather than an arbitrary or collapsed assignment. And modularity should matter only because it is the mechanism by which a model *composes* its specialized parts robustly when its environment changes – when the same task is presented under surface noise, paraphrase, or distribution shift.

This paper makes that three-step chain – consistency, for modularity, for robustness to a changing environment – explicit and measures each link directly, rather than treating routing consistency as a terminal objective. We build on CAR-MoE (Cache-Aware Routing for MoE), which fine-tunes a router with GRPO against a stateful reward: the hit ratio of a simulated fixed-capacity LRU cache of experts, so that the policy is rewarded for sequence-level expert reuse without any custom auxiliary loss or architectural change to the routing mechanism. We compare it against a plain base model, supervised fine-tuning (SFT) alone, an option-critic-style temporally extended controller [1, 2], and MELINOE [3], on two backbones of different scale (Phi-tiny-MoE-instruct: 16 experts, top-2; OLMoE-1B-7B: 64 experts, top-8), and ask, for each: is its routing consistent under meaning-preserving input perturbations; do its experts specialize by content domain; and does that consistency and specialization actually predict robustness on a downstream task under the same kind of perturbation. The answer is not uniformly yes, and the exception is informative: an objective can appear maximally consistent and modular by the aggregate statistics alone while being the least robust of the comparison, because it achieves that appearance through routing collapse rather than genuine specialization.

2 Method

CAR-MoE. We simulate a fixed-capacity LRU cache of experts at one middle transformer layer. Let $p_t \in \Delta^E$ be the router's distribution over E experts at token t and $\mathcal{C}_{t^-}$ the set of experts resident in the cache before token t, for a target capacity c. We use the soft reward $r_t = \sum_{e \in \mathcal{C}_{t^-}} p_t[e]$, the probability mass already captured by the working set, and $R(x, y) = \frac{1}{T}\sum_t r_t$ as the sequence-level reward for GRPO/DAPO [4, 5]. Every objective, baselines included, is a LoRA adapter over the frozen backbone, so absolute downstream scores are somewhat lower than a full fine-tune, uniformly across methods; we report benchmark performance as Δ from each backbone's own base model (Table 1), alongside the working-set hit ratio and mean cache misses from the same simulated LRU cache CAR-MoE is trained against.

Consistency. For a held-out prompt, we take a teacher-forced forward pass and pool the top-1 expert choice over all valid tokens into a per-expert usage histogram. We build two meaning-preserving perturbed copies of each prompt: character-level typo noise (5% per-character corruption rate) and cosmetic case/whitespace noise. Consistency is the Jensen-Shannon divergence between the original and perturbed histograms, averaged over 256 held-out prompts; lower divergence means the router's aggregate expert usage is invariant to surface form.

Modularity. We sample two disjoint-domain prompt pools (math and code, from the Nemotron Post-Training Dataset v2, 6) and compute each expert's domain-conditional usage $P(\text{math} \mid e) = \frac{n_{\text{math}}(e)}{n_{\text{math}}(e)+n_{\text{code}}(e)}$ from pooled usage counts over 256 prompts per domain. An expert's specialization index is $|P(\text{math} \mid e) - 0.5| \times 2 \in [0, 1]$; we report the mean specialization and the fraction of experts with specialization > 0.6. Good modularity should show up as invariance to the surface noise above *and* sensitivity to this genuine domain shift, the complement of the consistency measure.

Robustness. We build a small, self-contained GSM8K exact-match evaluation (regex `"#### <n>"` extraction, four few-shot exemplars, no lm-eval-harness) on 150 held-out test questions, once original and once perturbed with the same 5% typo noise as above, and report the accuracy drop as the robustness measure – a changing environment operationalized as surface noise, comparable to the consistency perturbation.

3 Results

Table 1: Benchmark performance, working-set efficiency, routing consistency, modularity, and robustness for every (backbone, objective) combination, at each backbone's trained working-set capacity (Phi: $c = 4$; OLMoE: $c = 16$). Avg ΔBench: mean Δ from the backbone's own base model over MMLU, MMMLU, GSM8K, HumanEval, and MATH (positive is better). Hit ratio / Misses: working-set hit ratio and mean misses per sequence from the same simulated LRU cache CAR-MoE is trained against (higher hit ratio / lower misses is better). JS-typo/JS-fmt: Jensen-Shannon divergence between original and perturbed routing histograms (lower = more consistent). Spec.-frac: fraction of experts with specialization > 0.6 (higher = more modular). ΔAcc: GSM8K accuracy drop under typo perturbation (lower = more robust). CAR-MoE rows highlighted; the option-critic controller's outlier row is marked † (Section 3.2).

Backbone	Objective	Avg ΔBench	Hit ratio	Misses	JS-typo	JS-fmt	Spec.-frac	ΔAcc
Phi-tiny-MoE	Base (untuned)	–	0.571	649.8	0.055	0.014	0.25	0.233
Phi-tiny-MoE	SFT	−0.01	0.565	742.1	0.053	0.014	0.25	0.247
Phi-tiny-MoE	CAR-MoE	−0.01	0.622	435.9	0.054	0.015	0.25	**0.220**
Phi-tiny-MoE	Option-critic†	−0.05	**1.000**	**0.1**	0.0001	0.00003	**0.94**	0.287
Phi-tiny-MoE	MELINOE	−0.05	0.879	237.7	0.046	0.011	0.44	0.333
OLMoE-1B-7B	Base (untuned)	–	0.686	1874.2	0.096	0.031	0.53	0.080
OLMoE-1B-7B	SFT	−0.05	0.679	1943.9	0.095	0.031	0.56	0.233
OLMoE-1B-7B	CAR-MoE	+0.01	**0.759**	**1004.2**	0.094	0.030	0.52	**0.073**
OLMoE-1B-7B	Option-critic	−0.11	0.684	2012.7	0.100	0.033	0.58	0.167
OLMoE-1B-7B	MELINOE	−0.14	0.763	1541.9	0.081	0.026	0.55	0.173

3.1 CAR-MoE is the most robust objective on both backbones

CAR-MoE achieves the smallest GSM8K accuracy drop of any objective tested, on *both* backbones: 0.073 on OLMoE-1B-7B (versus 0.080 base, 0.233 SFT – $3.2\times$ larger) and 0.220 on Phi-tiny-MoE (versus 0.233 base, 0.247 SFT, 0.333 MELINOE). On OLMoE-1B-7B it is the only objective that beats the untuned base outright, while also raising the hit ratio the most of any non-collapsed objective (0.759 vs. 0.686 base, a 46% cut in misses, 1004 vs. 1874) and losing the least average benchmark score (−0.01 vs. −0.05 SFT, −0.11 option-critic, −0.14 MELINOE) – efficiency, task performance, and robustness improve together rather than trading off. Phi-tiny-MoE shows the same pattern at smaller magnitude (hit ratio 0.622, misses 435.9, both better than base's 0.571/649.8, alongside the best robustness). It reaches all of this without any spike in aggregate consistency or specialization statistics – JS-typo, JS-fmt, and specialization sit within noise of base/SFT on the same backbone (Table 1). We read this as the reward improving robustness through genuine, non-collapsing reuse of an already-reasonable routing policy, not a conspicuous shift in aggregate behavior – unlike the mechanism discussed next.

3.2 The option-critic counterexample

Phi-tiny-MoE's option-critic controller breaks the pattern sharply. Its routing consistency is roughly three orders of magnitude better than every other objective on either backbone (JS-typo ≈ 0.0001 vs. 0.05–0.10 elsewhere), its specialization fraction (0.94) is the highest, and it reaches a perfect working-set hit ratio (1.000, 0.1 mean misses vs. hundreds elsewhere) – by every aggregate statistic in Table 1 this looks like the clearest success case for the consistency-to- modularity story and the best outcome for the underlying serving-efficiency motivation. It is instead the *least* robust objective tested ($\Delta\text{Acc} = 0.287$) and among the weakest on average benchmark score (-0.05). A routing trace analysis on the same checkpoint explains why: the controller collapses onto a small, fixed subset of experts resident for essentially the whole sequence rather than tracking content, so the perfect hit ratio is a symptom of that collapse, not evidence of useful reuse. A collapsed router is trivially perturbation-invariant (it barely uses its input to route at all) and trivially "specialized" by our domain statistic (a handful of experts absorb nearly all traffic from both domains, each nominally tied to whichever sent slightly more tokens). Neither statistic distinguishes this from genuine specialization without the robustness check: consistency and specialization are each consistent with meaningful modularity *or* degenerate collapse, and only a direct robustness measurement under the kind of environment change the

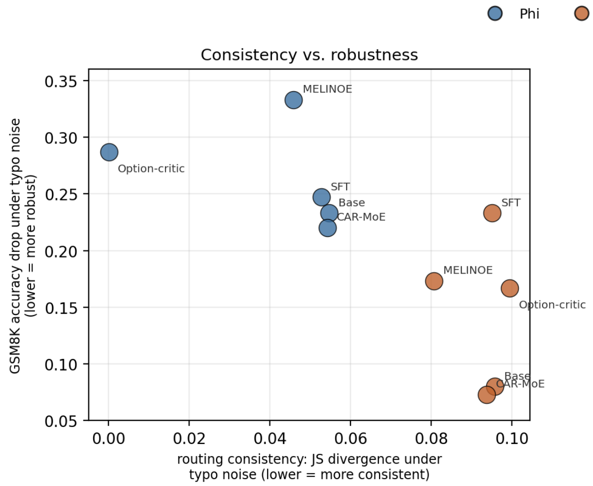

Figure 1: Left: consistency (JS divergence under typo noise) vs. GSM8K accuracy drop under the same perturbation. Right: modularity (fraction of specialized experts) vs. the same drop. CAR-MoE and Base/SFT cluster together within each backbone; Phi-tiny-MoE's option-critic controller (labeled) is the outlier in both panels – maximally consistent and "specialized," yet least robust.

modularity claim is about tells the two apart. Reporting consistency or hit ratio alone, as prior work in this space largely does, risks mistaking the latter for the former.

4 Conclusion

Routing consistency is not an end in itself; it is only useful evidence of modularity, and modularity is only useful insofar as it makes a model robust to a changing environment. Measuring all three links directly across two backbones and five objectives, we found CAR-MoE the most robust objective on both, together with the best working-set hit ratio and benchmark retention of any non-collapsed objective – while an option-critic controller reached the best aggregate consistency, specialization, *and* hit-ratio scores of any objective tested by collapsing its routing entirely, and paid for it with the worst downstream robustness in the comparison. Future work in this space should report a direct robustness check alongside any consistency or hit-ratio metric, since the latter, alone, cannot rule out this degenerate explanation.

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A Per-Expert Specialization Detail

Table 1's specialization fraction is the share of experts with $|P(\text{math} \mid e) - 0.5| \times 2 > 0.6$ out of E total experts (16 for Phi-tiny-MoE, 64 for OLMoE-1B-7B). The mean specialization index (not shown in the main table for space) follows the same ordering as the fraction reported: Phi-tiny-MoE base 0.450, SFT 0.434, CAR-MoE 0.430, option-critic 0.938, MELINOE 0.533; OLMoE base 0.563, SFT 0.612, CAR-MoE 0.567, option-critic 0.604, MELINOE 0.631. The cross-domain Jensen-Shannon divergence between each objective's math- and code-domain expert usage distributions – a complementary modularity signal, high values indicating the router genuinely reroutes between domains rather than using a fixed subset regardless of content – is: Phi-tiny-MoE base 0.159, SFT 0.138, CAR-MoE 0.149, option-critic 0.0001, MELINOE 0.243; OLMoE base 0.128, SFT 0.150, CAR-MoE 0.128, option-critic 0.139, MELINOE 0.121. Notably, the option-critic controller's near-zero cross-domain divergence confirms the collapse account directly: a router that used the same fixed experts regardless of domain would show zero divergence between domains by construction, exactly what is observed, whereas every other objective on either backbone shows a divergence at least three orders of magnitude larger.